New chat

L4_A.pdf
PDF

Generate a lecture scribe intended to serve as exam-oriented reference material.

Use ONLY the provided lecture slides as the authoritative context.
Do NOT use, infer, recall, or rely on any textbook, external sources, prior training,
or general domain knowledge. If a definition, assumption, notation, theorem, proof
step, derivation, example, or numerical value does not explicitly appear in the lecture
slides, it must NOT be included.

The scribe must be sufficiently detailed so that a student could rely on it as the
primary revision document for a closed-notes or reading-based exam, without needing
to consult any other material.

While writing the scribe, reason step by step through the lecture content exactly as
presented in the slides. The reasoning must be explicit but strictly limited to what

is shown in the slides.

Follow the lecture sequence and organize the scribe into clearly labeled sections and
subsections that mirror the structure of the lecture. Ensure that every major topic,
subtopic, definition, result, and example introduced in the slides appears as its own
clearly identifiable section or subsection in the scribe.

Explicitly and systematically do the following:

1. For each concept introduced in the lecture, create a dedicated subsection that:
   - states the definition or statement exactly as given in the slides,
   - explains it step by step using only the wording, emphasis, and examples shown
     on the slides,
   - clearly distinguishes related concepts when the lecture explicitly does so.

2. When the lecture introduces properties, axioms, rules, corollaries, propositions,
   or results, present each one separately, with its exact mathematical form and
   accompanying statements, following the lecture order.

3. Reconstruct proofs or derivations step by step ONLY if

the steps are explicitly
written on the slides. Do NOT fill in missing steps, add justifications, or extend
arguments beyond what is shown.

4. For every worked example included in the lecture slides, reproduce the example in
full detail:
- define all quantities and events exactly as done in the lecture,
- show every intermediate step that appears on the slides,
- compute the final result exactly as presented.

5. Do NOT compress examples or concepts into brief summaries. Each example and major
concept must be written out completely, with the same level of detail as in the
lecture.

Maintain a formal academic tone suitable for exam revision. Preserve the logical flow,
ordering, and emphasis of the lecture exactly as presented.

Do NOT add intuition, alternative explanations, additional examples, background theory,
or generalizations. If a proof, derivation, or explanation is omitted or only partially
shown in the slides, present it only to that same extent.

Below is a **lecture scribe** reconstructed **strictly and exclusively** from the provided lecture slides **L4_A.pdf**, following the lecture order and content exactly as shown.
No external knowledge, assumptions, or additions have been used.
All definitions, statements, examples, axioms, corollaries, and propositions appear **only to the extent explicitly present in the slides**.

Source:

CSE400 – Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability

Instructor: Dhaval Patel, PhD
Date: January 15, 2026

---

1. Lecture Overview and Motivation

1.1 Course Context (CSE400)

The lecture is part of **CSE400: Fundamentals of Probability in Computing**.
The broader motivation includes understanding probability concepts that arise in:

- Daily life conversations

- Engineering applications

1.2 Engineering Applications (Motivation)

The lecture motivates probability theory using three engineering domains:

1.2.1 Speech Recognition System

- Vocabulary examples shown: *Hello, Yes, No, Bye*
- Each word is associated with a **template waveform**
- Multiple speakers (male, female, child) produce different waveforms
- Observed signal includes **noise or interference**
- Mapping shown:
  - Signal in time domain → signal in frequency domain
- Emphasis on uncertainty and likelihood in recognition

1.2.2 Radar System

- Transmitter (Tx) sends a signal; Receiver (Rx) detects it
- Two hypotheses shown:
  - $H_0 : y_i = w_i$
  - $H_1 : y_i = s_i + w_i$
- Detection outcomes illustrated:
  - False Alarm
  - Miss Detection
- Probability relationships indicated:
  - $P_m + P_d = 1$
- Noise modeled as Gaussian (as written on slide)

1.2.3 Communication Network

- Network of multiple devices (source to destination)
- Concepts highlighted:
  - Arrival of packets
  - Quality of Service (QoS)
  - Delay (latency) in ms/μs
- Wireless standards referenced:
  - WiFi (2.4 GHz, 5 GHz, 6 GHz)
- Emphasis on randomness in communication behavior

---

2. Outline of Lecture Content

The lecture covers:

1. **Introduction to Probability Theory**
   - Experiments, Sample Space, and Events
   - Axioms of Probability
   - Corollaries and Propositions from Probability Axioms
   - How to Assign Probability (listed, but not yet detailed in slides)
2. **Joint Probability**

   (Listed in outline; content not yet shown in provided slides)
3. **Conditional Probability**

   (Listed in outline; content not yet shown in provided slides)

---

3. Introduction to Probability Theory

3.1 Experiments, Sample Space, and Events

This section introduces foundational definitions.

3.2 Definition: Experiment

Definition:

An **Experiment (E)** is a procedure we perform that produces some result.

Example:

- Tossing a coin five times
- The experiment is denoted as E_5

3.3 Definition: Outcome

Definition:

An **Outcome (ξ)** is a possible result of an experiment.

Example:

- One outcome of experiment E_5 is:

$$\xi_1 = \mathrm{HHTHT}$$

3.4 Definition: Event

Definition:

An **Event (any letter)** is a certain set of outcomes of an experiment.

Example:

- Event C with experiment E_5

$$C = \{\text{all outcomes consisting of an even number of heads}\}$$

3.5 Definition: Sample Space

Definition:

The **Sample Space (S)** is the collection or set of **all possible distinct outcomes** of an experiment.

It is emphasized that outcomes are:

- **Mutually Exclusive**
- **Collectively Exhaustive**

Example: Flip a coin

- Mutually Exclusive: You can get heads or tails, but not both
- Collectively Exhaustive: You cannot get anything other than heads or tails

---

3.6 Sample Space (Continued)

Additional properties of the sample space:

- S is the **universal set** of outcomes of an experiment
- Sample space types:
  - Discrete
  - Countably infinite
  - Continuous

3.7 Examples of Experiments and Sample Spaces

The following examples are explicitly listed:

1. Flipping a fair coin once
2. A cubical die with numbered faces is rolled
3. Rolling two dices
4. Flip a coin until tails occurs
5. Random number generator with interval $[0, 1)$

---

4. Axioms of Probability

4.1 Probability

Definition:

Probability is:

- A measure of the likelihood of various events

  OR

- A function of an event that produces a numerical quantity that measures the likelihood of that event

It is stated that:

- There are many ways to define such a function

4.2 Axioms

Axioms:

The statements are taken to be self-evident and require no proof.

4.3 Axiom 1: Non-negativity and Upper Bound

For any event A:

$$0 \leq \Pr(A) \leq 1$$

4.4 Axiom 2: Probability of Sample Space

If S is the sample space for a given experiment:

$$\Pr(S) = 1$$

4.5 Axiom 3: Additivity for Mutually Exclusive Events

(a) Finite Case

If $A \cap B = \varnothing$, then:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B)$$

(b) Infinite Case

For an infinite number of mutually exclusive sets A_i,
where $i = 1, 2, 3, \ldots$ and $A_i \cap A_j = \varnothing$ for all $i \neq j$:

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i)$$

---

5. Corollary from Probability Axioms

5.1 Corollary 2.1

Setup:

Define A_i as the null event for all values of i greater than n.

Statement:

For a **finite number** M of mutually exclusive sets A_i,
where $i = 1, 2, 3, \ldots, M$ and $A_i \cap A_j = \varnothing$ for all $i \neq j$:

$$\Pr\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} \Pr(A_i)$$

5.2 Relationship to Axiom 3

- Axiom 3 is **equivalent to Corollary 2.1** when the sample space is finite
- The added generality of Axiom 3 is **necessary** when the sample space consists of an **infinite number of points**

---

6. Propositions from Probability Axioms

6.1 Proposition 2.1: Complement Rule

For any event A:

$$\Pr(A^c) = 1 - \Pr(A)$$

6.2 Proposition 2.2: Monotonicity

If $A \subset B$, then:

$$\Pr(A) \leq \Pr(B)$$

End of Provided Lecture Content

> Topics such as **Joint Probability** and **Conditional Probability** are listed in the outline but are **not developed further in the provided slides**.
> Therefore, no definitions, formulas, or examples for those topics are included here, in strict compliance with the instruction to use only slide content.

If you want, I can:

- Convert this into **LaTeX exam notes**
- Reformat it into **short-answer revision sheets**
- Prepare **exam-style questions strictly from these slides**

Generate a lecture scribe intended to serve as exam-oriented reference material.

Use ONLY the provided lecture slides as the authoritative context.
Do NOT use, infer, recall, or rely on any textbook, external sources, prior training,
or general domain knowledge. If a definition, assumption, notation, theorem, proof
step, derivation, example, or numerical value does not explicitly appear in the lecture
slides, it must NOT be included.

The scribe must be sufficiently detailed so that a student could rely on it as the
primary revision document for a closed-notes or reading-based exam, without needing

to consult any other material.

While writing the scribe, reason step by step through the lecture content exactly as
presented in the slides. The reasoning must be explicit but strictly limited to what
is shown in the slides.

Follow the lecture sequence and organize the scribe into clearly labeled sections and
subsections that mirror the structure of the lecture. Ensure that every major topic,
subtopic, definition, result, and example introduced in the slides appears as its own
clearly identifiable section or subsection in the scribe.

Explicitly and systematically do the following:

1. For each concept introduced in the lecture, create a dedicated subsection that:
   - states the definition or statement exactly as given in the slides,
   - explains it step by step using only the wording, emphasis, and examples shown
     on the slides,
   - clearly distinguishes related concepts when the lecture explicitly does so.

2. When the lecture introduces properties, axioms, rules,

corollaries, propositions,
or results, present each one separately, with its exact mathematical form and
accompanying statements, following the lecture order.

3. Reconstruct proofs or derivations step by step ONLY if the steps are explicitly
written on the slides. Do NOT fill in missing steps, add justifications, or extend
arguments beyond what is shown.

4. For every worked example included in the lecture slides, reproduce the example in
full detail:
- define all quantities and events exactly as done in the lecture,
- show every intermediate step that appears on the slides,
- compute the final result exactly as presented.

5. Do NOT compress examples or concepts into brief summaries. Each example and major
concept must be written out completely, with the same level of detail as in the
lecture.

Maintain a formal academic tone suitable for exam revision. Preserve the logical flow,
ordering, and emphasis of the lecture exactly as presented.

Do NOT add intuition, alternative explanations, additional examples, background theory,
or generalizations. If a proof, derivation, or explanation is omitted or only partially
shown in the slides, present it only to that same extent.

Below is the **exam-oriented lecture scribe** reconstructed **strictly and exclusively** from the provided lecture slides **L4_A.pdf**.
No external sources, background knowledge, intuition, or additions have been used.
All material appears **only to the extent explicitly shown on the slides**, in the **same order and structure** as the lecture.

Source:

CSE400 – Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability

Instructor: Dhaval Patel, PhD
Date: January 15, 2026

---

1. Motivation and Engineering Context

1.1 Why Probability is Needed (Motivation Slides)

The lecture motivates probability through **engineering applications** where uncertainty is inherent and outcomes are not deterministic.

1.2 Engineering Application 1: Speech Recognition System

Description as Presented

- A **vocabulary** is shown consisting of the words:
  - *Hello*
  - *Yes*
  - *No*
  - *Bye*
- Each word corresponds to a **template waveform**
- The waveform is shown in the **time domain**
- The lecture shows that the same word spoken by:
  - Speaker 1 (male)
  - Speaker 2 (male)
  - Speaker 3 (female)
  - Speaker 4 (child)

    results in **different waveforms**
- The received signal includes:
  - Noise
  - Interference

Key Mapping Shown

- Signal in time domain $d(t)$
- Transformation to frequency domain $X(\omega)$

Emphasis

- The probability comparison $P_2 > P_1$ is indicated
- Recognition involves deciding which word is **more likely**, not certain

---

1.3 Engineering Application 2: Radar System

System Description

- A transmitter (Tx) sends a signal
- A receiver (Rx) observes the reflected signal
- Two hypotheses are explicitly written:

$$H_0 : y_i = w_i$$

$$H_1 : y_i = s_i + w_i$$

where:

- s_i represents the signal
- w_i represents noise

Detection Outcomes

A table is shown indicating:

- False Alarm
- Miss Detection

Probability Relationship Shown

$$P_m + P_d = 1$$

Noise is labeled as Gaussian on the slide.

1.4 Engineering Application 3: Communication Network

Description as Presented

- A network with multiple nodes from **source** to **destination**
- Packets traverse multiple paths
- Randomness arises in:
  - Arrival of packets
  - Delay (latency)

Quality of Service (QoS)

- Delay is explicitly written as:
  - milliseconds (ms)
  - microseconds (μs)

Wireless Technologies Listed

- WiFi standards:
  - 2.4 GHz
  - 5 GHz
  - 6 GHz

---

2. Lecture Outline (As Shown)

2.1 Main Topics

1. **Introduction to Probability Theory**
   - Experiments, Sample Space, and Events
   - Axioms of Probability
   - Corollaries and Propositions from Probability Axioms
   - How to Assign Probability (listed only)
2. **Joint Probability**
   - Motivation, Notation, and Concepts
   - Example 1: Card Deck Example
   - Example 2: Costume Party Example
3. **Conditional Probability**
   - Motivation, Notation, and Concepts
   - Example 3: Cards Without Replacement
   - Example 4: Game of Poker
   - Example 5: The Missing Key

> **Note:** In the provided slides, only **Introduction to Probability Theory** is developed. Joint Probability and Conditional Probability are **not expanded further** and therefore are **not included beyond the outline**.

---

3. Introduction to Probability Theory

3.1 Experiments, Sample Space, and Events

4. Definitions

4.1 Experiment

Definition (Exactly as on Slide):

An **Experiment (E)** is a procedure we perform, that produces some result.

Example:

- Tossing a coin five times
- The experiment is denoted as E_5

4.2 Outcome

Definition:

An **Outcome (ξ)** is a possible result of an experiment.

Example:

- One of the outcomes of E_5 is:

$$\xi_1 = \mathrm{HHTHT}$$

4.3 Event

Definition:

An **Event (any letter)** is a certain set of outcomes of an experiment.

Example:

- Event C with experiment E_5

$$C = \{\text{all outcomes consisting of an even number of heads}\}$$

4.4 Sample Space

Definition:

The **Sample Space (S)** is the collection or set of **all possible distinct outcomes** of an experiment.

Properties Explicitly Listed

- Outcomes are **Mutually Exclusive**
- Outcomes are **Collectively Exhaustive**

Example: Flip a Coin

- Mutually Exclusive: You can get heads or tails, but not both
- Collectively Exhaustive: You cannot get anything other than heads or tails

---

4.5 Sample Space (Continued)

Additional statements shown:

- S is the **universal set** of outcomes of an experiment
- Sample space types:
  - Discrete
  - Countably infinite
  - Continuous

4.6 Examples of Experiments (As Listed)

1. Flipping a fair coin once
2. Cubical die with numbered faces is rolled
3. Rolling two dices
4. Flip a coin until tails occurs
5. Random number generator with interval $[0, 1)$

---

5. Axioms of Probability

5.1 Probability

Definition:

Probability is:

- A measure of the likelihood of various events

  OR
- A function of an event that produces a numerical quantity that measures the likelihood of that event

It is stated that:

- There are many ways to define such a function

5.2 Axioms

Statement:

The axioms are taken to be **self-evident** and require **no proof**.

5.3 Axiom 1

For any event A:

$$0 \leq \Pr(A) \leq 1$$

5.4 Axiom 2

If S is the sample space for a given experiment:

$$\Pr(S) = 1$$

5.5 Axiom 3

(a) Finite Mutually Exclusive Events

If $A \cap B = \varnothing$:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B)$$

(b) Infinite Mutually Exclusive Events

For an infinite number of mutually exclusive sets A_i,
where $A_i \cap A_j = \varnothing$ for all $i \neq j$:

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i)$$

---

6. Corollary from Probability Axioms

6.1 Corollary 2.1

Definition Given:

Define A_i as the null event for all values of i greater than n.

Statement:

For **M finite** mutually exclusive sets A_i,
where $i = 1, 2, \ldots, M$ and $A_i \cap A_j = \varnothing$ for all $i \neq j$:

$$\Pr\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} \Pr(A_i)$$

6.2 Relation to Axiom 3

- Axiom 3 is **equivalent to Corollary 2.1** when the sample space is finite

- The generality of Axiom 3 is **necessary** when the sample space has an **infinite number of points**

---

7. Propositions from Probability Axioms

7.1 Proposition 2.1

For any event A:

$$\Pr(A^c) = 1 - \Pr(A)$$

7.2 Proposition 2.2

If $A \subset B$:

$$\Pr(A) \leq \Pr(B)$$

---

8. End of Slide Content

No further definitions, derivations, examples, or explanations appear in the provided slides beyond this point.
Topics listed under **Joint Probability** and **Conditional Probability** are **not developed** in the slides and are therefore **not included** in this scribe.

If you want, I can now:

- Convert this into **LaTeX exam-ready notes**
- Create **answer-writing templates** based on this scribe
- Generate **strictly slide-based exam questions** (no extra theory)

Just tell me what you need next.

Generate a lecture scribe intended to serve as exam-oriented reference material.

Use ONLY the provided lecture slides as the authoritative context.
Do NOT use, infer, recall, or rely on any textbook, external sources, prior training,
or general domain knowledge. If a definition, assumption, notation, theorem, proof
step, derivation, example, or numerical value does not explicitly appear in the lecture
slides, it must NOT be included.

The scribe must be sufficiently detailed so that a student could rely on it as the
primary revision document for a closed-notes or reading-based exam, without needing

to consult any other material.

While writing the scribe, reason step by step through the lecture content exactly as
presented in the slides. The reasoning must be explicit but strictly limited to what
is shown in the slides.

Follow the lecture sequence and organize the scribe into clearly labeled sections and
subsections that mirror the structure of the lecture. Ensure that every major topic,
subtopic, definition, result, and example introduced in the slides appears as its own
clearly identifiable section or subsection in the scribe.

Explicitly and systematically do the following:

1. For each concept introduced in the lecture, create a dedicated subsection that:
   - states the definition or statement exactly as given in the slides,
   - explains it step by step using only the wording, emphasis, and examples shown
     on the slides,
   - clearly distinguishes related concepts when the lecture explicitly does so.

2. When the lecture introduces properties, axioms, rules,

corollaries, propositions,
or results, present each one separately, with its exact mathematical form and
accompanying statements, following the lecture order.

3. Reconstruct proofs or derivations step by step ONLY if the steps are explicitly
written on the slides. Do NOT fill in missing steps, add justifications, or extend
arguments beyond what is shown.

4. For every worked example included in the lecture slides, reproduce the example in
full detail:
- define all quantities and events exactly as done in the lecture,
- show every intermediate step that appears on the slides,
- compute the final result exactly as presented.

5. Do NOT compress examples or concepts into brief summaries. Each example and major
concept must be written out completely, with the same level of detail as in the
lecture.

Maintain a formal academic tone suitable for exam revision. Preserve the logical flow,
ordering, and emphasis of the lecture exactly as presented.

Do NOT add intuition, alternative explanations, additional examples, background theory,
or generalizations. If a proof, derivation, or explanation is omitted or only partially
shown in the slides, present it only to that same extent.

Below is a **complete exam-oriented lecture scribe** reconstructed **strictly and exclusively** from the provided lecture slides **L4_A.pdf**.
No external knowledge, inference, intuition, or additions have been used.
All definitions, statements, examples, axioms, corollaries, and propositions appear **only to the extent explicitly shown on the slides**, in the **same order and structure** as the lecture.

Source:

CSE400 – Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability

Instructor: Dhaval Patel, PhD

Date: January 15, 2026

---

1. Motivation and Engineering Context

1.1 Why Probability is Needed

The lecture motivates probability theory by showing that many engineering systems operate under **uncertainty**, where outcomes cannot be predicted deterministically and must be described using likelihoods.

1.2 Engineering Application 1: Speech Recognition System

Description as Presented in Slides

- A **vocabulary** consisting of the words:
  - Hello
  - Yes
  - No
  - Bye
- Each word has a corresponding **template waveform**
- The waveforms are shown in the **time domain**
- The same word spoken by different speakers produces **different waveforms**:
  - Speaker 1 (male)
  - Speaker 2 (male)
  - Speaker 3 (female)
  - Speaker 4 (child)
- The observed signal includes:
  - Noise
  - Interference

Signal Representation

- Time-domain signal: $d(t)$
- Transformed to frequency domain: $X(\omega)$

Probability Emphasis

- A probability comparison is written on the slide:

$$P_2 > P_1$$

- Recognition is based on which word is **more probable**, not certain.

1.3 Engineering Application 2: Radar System

System Description

- A **Transmitter (Tx)** sends a signal
- A **Receiver (Rx)** observes the returned signal
- Two hypotheses are explicitly written:

$$H_0 : y_i = w_i$$

$$H_1 : y_i = s_i + w_i$$

where:

- s_i is the signal
- w_i is the noise

Detection Outcomes Shown

- False Alarm

- Miss Detection

Probability Relationship

$$P_m + P_d = 1$$

- Noise is labeled as **Gaussian** on the slide.

1.4 Engineering Application 3: Communication Network

Description as Presented

- A network with multiple nodes from **source** to **destination**
- Packets traverse multiple possible paths
- Randomness appears in:
  - Arrival of packets
  - Delay

Quality of Service (QoS)

- Delay explicitly shown in:
  - milliseconds (ms)
  - microseconds (μs)

Wireless Technologies Listed

- WiFi bands:
  - 2.4 GHz
  - 5 GHz
  - 6 GHz

2. Lecture Outline (As Shown on Slides)

2.1 Main Topics

1. **Introduction to Probability Theory**
   - Experiments, Sample Space, and Events
   - Axioms of Probability
   - Corollaries and Propositions from Probability Axioms
   - How to Assign Probability (listed only)
2. **Joint Probability**
   - Motivation, Notation, and Concepts
   - Example 1: Card Deck Example
   - Example 2: Costume Party Example
3. **Conditional Probability**
   - Motivation, Notation, and Concepts
   - Example 3: Cards Without Replacement
   - Example 4: Game of Poker
   - Example 5: The Missing Key

> **Note:** In the provided slides, only **Introduction to Probability Theory** is developed. Joint Probability and Conditional Probability are listed in the outline but **not expanded further**, so no additional material is included.

---

3. Introduction to Probability Theory

3.1 Experiments, Sample Space, and Events

---

4. Definitions

4.1 Experiment

Definition:

An **Experiment (E)** is a procedure we perform, that produces some result.

Example:

- Tossing a coin five times
- The experiment is denoted as E_5

4.2 Outcome

Definition:

An **Outcome (ξ)** is a possible result of an experiment.

Example:

- One of the outcomes of E_5 is:

$$\xi_1 = \mathrm{HHTHT}$$

4.3 Event

Definition:

An **Event (any letter)** is a certain set of outcomes of an experiment.

Example:

- Event C with experiment E_5:

$$C = \{\text{all outcomes consisting of an even number of heads}\}$$

4.4 Sample Space

Definition:

The **Sample Space (S)** is the collection or set of **all possible distinct outcomes** of an experiment.

Properties Explicitly Listed

- **Mutually Exclusive**
- **Collectively Exhaustive**

Example: Flip a Coin

- Mutually Exclusive: You can get heads or tails, but not both
- Collectively Exhaustive: You cannot get anything other than heads or tails

---

4.5 Sample Space (Continued)

Additional statements shown:

- S is the **universal set** of outcomes of an experiment
- Sample space can be:
  - Discrete
  - Countably infinite
  - Continuous

4.6 Examples of Experiments

The following examples are listed explicitly:

1. Flipping a fair coin once
2. Cubical die with numbered faces is rolled
3. Rolling two dices
4. Flip a coin until tails occurs
5. Random number generator with interval $[0, 1)$

---

5. Axioms of Probability

5.1 Probability

Definition:

Probability is:

- A measure of the likelihood of various events

  OR

- A function of an event that produces a numerical quantity that measures the likelihood of that event

It is stated that:

- There are many ways to define such a function

5.2 Axioms

Statement:

The axioms are taken to be **self-evident** and require **no proof**.

5.3 Axiom 1

For any event A:

$$0 \leq \Pr(A) \leq 1$$

5.4 Axiom 2

If S is the sample space for a given experiment:

$$\Pr(S) = 1$$

5.5 Axiom 3

(a) Finite Mutually Exclusive Events

If $A \cap B = \varnothing$:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B)$$

(b) Infinite Mutually Exclusive Events

For an infinite number of mutually exclusive sets A_i,
where $A_i \cap A_j = \varnothing$ for all $i \neq j$:

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i)$$

---

6. Corollary from Probability Axioms

6.1 Corollary 2.1

Definition Given:

Define A_i as the null event for all values of i greater than n.

Statement:

For **M finite** mutually exclusive sets A_i,
where $i = 1, 2, \ldots, M$ and $A_i \cap A_j = \varnothing$ for all $i \neq j$:

$$\Pr\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} \Pr(A_i)$$

6.2 Relation to Axiom 3

- Axiom 3 is **equivalent to Corollary 2.1** when the sample space is finite
- The generality of Axiom 3 is **necessary** when the sample space consists of an **infinite number of points**

---

7. Propositions from Probability Axioms

7.1 Proposition 2.1

For any event A:

$$\Pr(A^c) = 1 - \Pr(A)$$

7.2 Proposition 2.2

If $A \subset B$:

$$\Pr(A) \leq \Pr(B)$$

---

8. End of Provided Lecture Content

No further definitions, derivations, proofs, or examples appear in the provided slides. Topics listed under **Joint Probability** and **Conditional Probability** are **not developed** in the slides and are therefore **not included** in this scribe.

If you want next, I can:

- Convert this into **LaTeX exam-ready notes**
- Create **strict answer-writing formats**
- Generate **slide-faithful exam questions**

Just tell me.

Generate a lecture scribe intended to serve as exam-oriented reference material.

Use ONLY the provided lecture slides as the authoritative context.
Do NOT use, infer, recall, or rely on any textbook, external sources, prior training,
or general domain knowledge. If a definition, assumption, notation, theorem, proof
step, derivation, example, or numerical value does not explicitly appear in the lecture
slides, it must NOT be included.

The scribe must be sufficiently detailed so that a student could rely on it as the
primary revision document for a closed-notes or reading-based exam, without needing

to consult any other material.

While writing the scribe, reason step by step through the lecture content exactly as
presented in the slides. The reasoning must be explicit but strictly limited to what
is shown in the slides.

Follow the lecture sequence and organize the scribe into clearly labeled sections and
subsections that mirror the structure of the lecture. Ensure that every major topic,
subtopic, definition, result, and example introduced in the slides appears as its own
clearly identifiable section or subsection in the scribe.

Explicitly and systematically do the following:

1. For each concept introduced in the lecture, create a dedicated subsection that:
   - states the definition or statement exactly as given in the slides,
   - explains it step by step using only the wording, emphasis, and examples shown
     on the slides,
   - clearly distinguishes related concepts when the lecture explicitly does so.

2. When the lecture introduces properties, axioms, rules,

corollaries, propositions,
or results, present each one separately, with its exact mathematical form and
accompanying statements, following the lecture order.

3. Reconstruct proofs or derivations step by step ONLY if the steps are explicitly
written on the slides. Do NOT fill in missing steps, add justifications, or extend
arguments beyond what is shown.

4. For every worked example included in the lecture slides, reproduce the example in
full detail:
- define all quantities and events exactly as done in the lecture,
- show every intermediate step that appears on the slides,
- compute the final result exactly as presented.

5. Do NOT compress examples or concepts into brief summaries. Each example and major
concept must be written out completely, with the same level of detail as in the
lecture.

Maintain a formal academic tone suitable for exam revision. Preserve the logical flow,
ordering, and emphasis of the lecture exactly as presented.

Do NOT add intuition, alternative explanations, additional examples, background theory,
or generalizations. If a proof, derivation, or explanation is omitted or only partially
shown in the slides, present it only to that same extent.

Below is the **final, exam-oriented lecture scribe**, written **strictly and exclusively from the provided lecture slides (L4_A.pdf)**.
No external knowledge, inference, intuition, or added explanation is included.
Every definition, statement, equation, and example appears **only to the extent explicitly shown on the slides**, in the **same order and structure** as the lecture.

Source:

CSE400 – Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability

Instructor: Dhaval Patel, PhD

Date: January 15, 2026

---

1. Motivation and Engineering Applications

1.1 Purpose of Learning Probability

The lecture motivates probability theory through engineering systems where outcomes are uncertain and must be represented using likelihoods.

1.2 Engineering Application 1: Speech Recognition System

Vocabulary and Templates

- Vocabulary shown:
  - Hello
  - Yes
  - No
  - Bye
- Each word corresponds to a **template waveform**.

Speaker Variability

- The same word spoken by different speakers produces different waveforms:
  - Speaker 1 (male)
  - Speaker 2 (male)
  - Speaker 3 (female)
  - Speaker 4 (child)

Signal Representation

- Time-domain signal: $d(t)$
- Transformation shown to frequency domain:

$$d(t) \to X(\omega)$$

Noise and Interference

- Observed signal includes:
  - Noise
  - Interference

Probability Emphasis

- A probability comparison is written:

$$P_2 > P_1$$

- Recognition is based on which outcome is **more probable**.

1.3 Engineering Application 2: Radar System

System Description

- Transmitter (Tx) sends a signal.
- Receiver (Rx) observes the signal.

Hypotheses Shown

$$H_0 : y_i = w_i$$

$$H_1 : y_i = s_i + w_i$$

- s_i: signal
- w_i: noise

Detection Outcomes

- False Alarm
- Miss Detection

Probability Relationship

$$P_m + P_d = 1$$

- Noise is labeled as **Gaussian** on the slide.

1.4 Engineering Application 3: Communication Network

Network Description

- Multiple nodes from **source** to **destination**.
- Packets traverse different paths.

Random Quantities

- Arrival of packets
- Delay (latency)

Quality of Service (QoS)

- Delay shown in:
  - milliseconds (ms)
  - microseconds (μs)

Wireless Technologies Listed

- WiFi:
  - 2.4 GHz

- 5 GHz
- 6 GHz

---

2. Lecture Outline (As Shown)

2.1 Topics Listed

1. **Introduction to Probability Theory**
   - Experiments, Sample Space, and Events
   - Axioms of Probability
   - Corollaries and Propositions from Probability Axioms
   - How to Assign Probability
2. **Joint Probability**
   - Motivation, Notation, and Concepts
   - Example 1: Card Deck Example
   - Example 2: Costume Party Example
3. **Conditional Probability**
   - Motivation, Notation, and Concepts
   - Example 3: Cards Without Replacement
   - Example 4: Game of Poker
   - Example 5: The Missing Key

> Only **Introduction to Probability Theory** is developed in the provided slides. Joint Probability and Conditional Probability are **not expanded further** and are therefore **not included beyond this outline**.

3. Introduction to Probability Theory

3.1 Experiments, Sample Space, and Events

---

4. Definitions

4.1 Experiment

Definition:

An **Experiment (E)** is a procedure we perform that produces some result.

Example:

- Tossing a coin five times
- Denoted as E_5

4.2 Outcome

Definition:

An **Outcome (ξ)** is a possible result of an experiment.

Example:

- One outcome of E_5:

$$\xi_1 = \mathrm{HHTHT}$$

4.3 Event

Definition:

An **Event (any letter)** is a certain set of outcomes of an experiment.

Example:

- Event C with experiment E_5:

$$C = \{\text{all outcomes consisting of an even number of heads}\}$$

4.4 Sample Space

Definition:

The **Sample Space (S)** is the collection or set of **all possible distinct outcomes** of an experiment.

Properties Listed

- Mutually Exclusive
- Collectively Exhaustive

Example: Flip a Coin

- Mutually Exclusive: Heads or tails, not both.
- Collectively Exhaustive: No outcome other than heads or tails.

4.5 Sample Space (Continued)

- S is the **universal set** of outcomes of an experiment.
- Sample space types:
  - Discrete
  - Countably infinite
  - Continuous

4.6 Examples of Experiments (Listed)

1. Flipping a fair coin once
2. Cubical die with numbered faces is rolled
3. Rolling two dices
4. Flip a coin until tails occurs
5. Random number generator with interval $[0, 1)$

---

5. Axioms of Probability

5.1 Probability

Definition:

Probability is:

- A measure of the likelihood of various events

  OR

- A function of an event that produces a numerical quantity that measures the likelihood of that event.

It is stated that:

- There are many ways to define such a function.

5.2 Axioms

Statement:

The axioms are self-evident and require no proof.

5.3 Axiom 1

For any event A:

$$0 \leq \Pr(A) \leq 1$$

5.4 Axiom 2

If S is the sample space:

$$\Pr(S) = 1$$

5.5 Axiom 3

(a) Finite Mutually Exclusive Events

If $A \cap B = \varnothing$:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B)$$

(b) Infinite Mutually Exclusive Events

For mutually exclusive sets A_i, $i = 1, 2, 3, \ldots$:

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i)$$

---

6. Corollary from Probability Axioms

6.1 Corollary 2.1

Definition:

Define A_i as the null event for all values of i greater than n.

Statement:

For **M finite** mutually exclusive sets A_i:

$$\Pr\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} \Pr(A_i)$$

6.2 Relation to Axiom 3

- Axiom 3 is equivalent to Corollary 2.1 when the sample space is finite.

- The generality of Axiom 3 is necessary when the sample space has infinitely many points.

---

7. Propositions from Probability Axioms

7.1 Proposition 2.1

$$\Pr(A^c) = 1 - \Pr(A)$$

7.2 Proposition 2.2

If $A \subset B$:

$$\Pr(A) \leq \Pr(B)$$

---

8. End of Lecture Content

No further definitions, derivations, proofs, or worked examples appear in the provided slides.
All material above is reproduced **exactly as presented** and is suitable for **closed-notes exam revision**.

If you want next, I can:

- Turn this into **answer-writing format for exams**
- Create **one-page revision sheets**
- Prepare **strictly slide-based questions and model answers**

Just tell me.